

Logistic Regression

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학습 내용

- 이진 분류가 무엇인지 설명할 수 있다.
- Logistic regression이 무엇인지 설명할 수 있다.

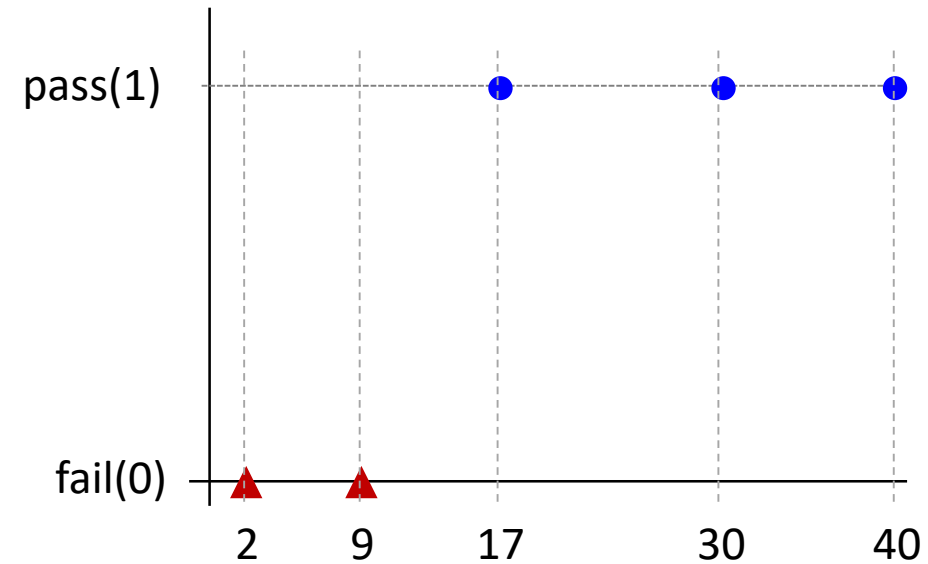
이진 분류 binary classification

- 전자메일^{email}: 스팸^{spam}(1) 또는 정상^{ham}(0)
- 전자결제^{online transaction}: 의심^{fraud}(1) 또는 정당^{legitimate}(0)
- 암^{tumor}: 악성^{malignant}(1) 또는 양성^{benign}(0)

$y \in \{0,1\}$ 0: Negative class
 1: Positive class

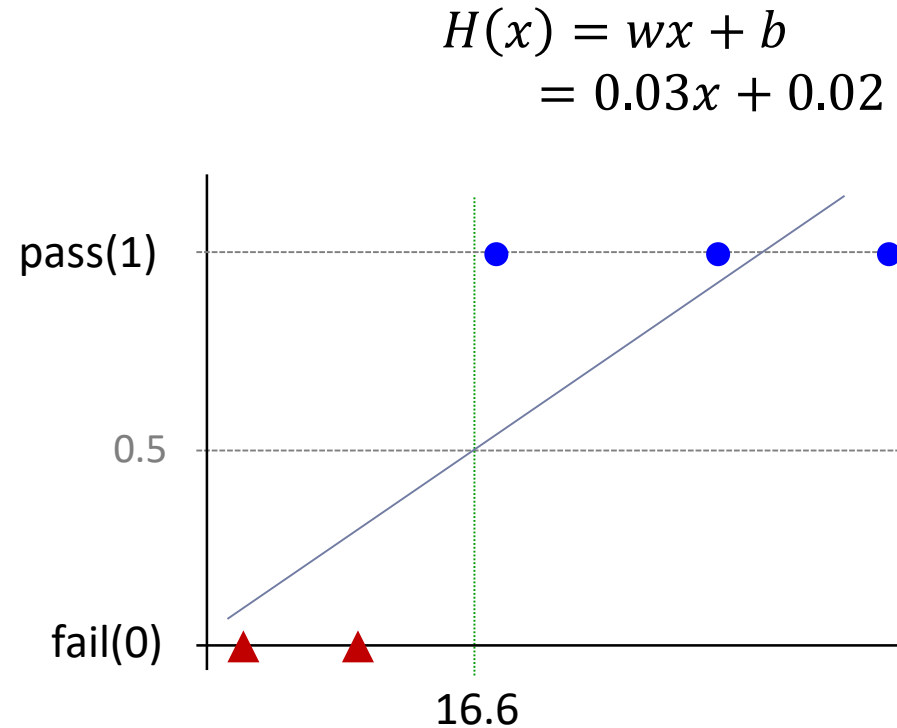
예제: 공부 시간에 따른 과목 이수 여부

공부 시간	과목 이수
9	fail(0)
30	pass(1)
17	pass(1)
2	fail(0)
40	pass(1)



Linear regression을 사용하면?

공부 시간	과목 이수
9	fail(0)
30	pass(1)
17	pass(1)
2	fail(0)
40	pass(1)

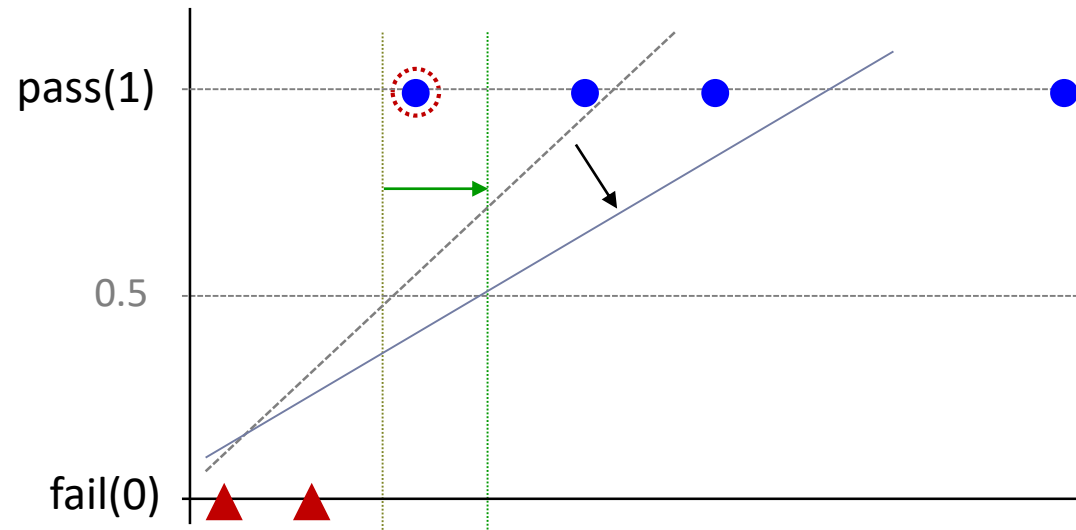


만약 $H(x) \geq 0.5$ 이면 $y = 1$

만약 $H(x) < 0.5$ 이면 $y = 0$

Linear regression의 어려움 (1)

공부 시간	과목 이수
9	fail(0)
30	pass(1)
17	pass(1)
2	fail(0)
40	pass(1)
90	pass(1)



Linear regression의 어려움 (2)

$$H(x) = wx + b$$

Binary classification: $y = 0$ 또는 $y = 1$

$H(x) \gg 1$ 또는 $H(x) \ll 0$ 가능

이진 분류의 경우 x 의 값에 따라 y 는 0 또는 1의 값만을 가져야 하지만
linear regression에서는 x 의 값이 매우 크거나 작을 경우에는
 y 도 매우 크거나 작은 값을 갖게 됨

Logistic Regression: $0 < H(x) < 1$

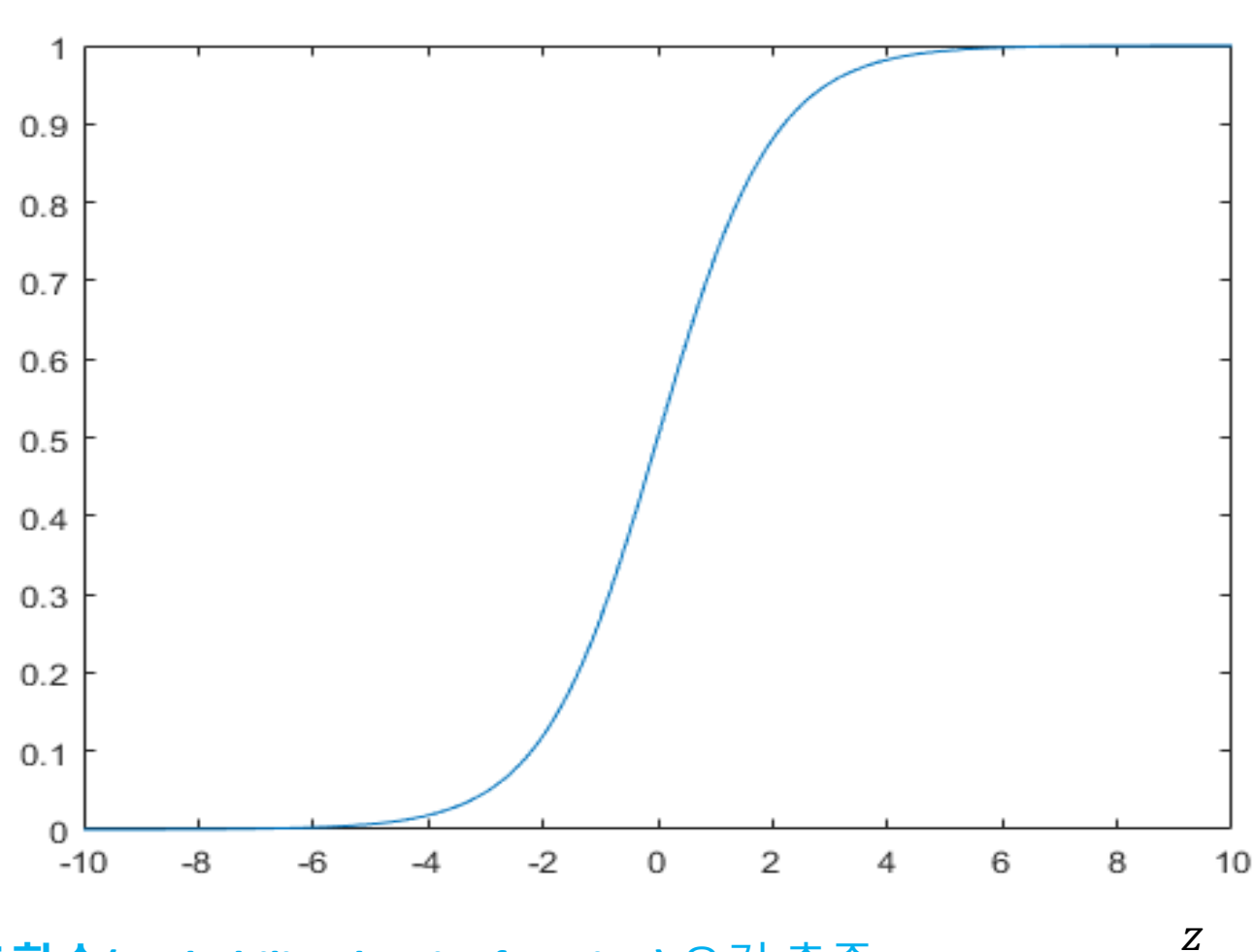
sigmoid function

Logistic function

$$\sigma(z) = \frac{L}{1 + e^{-k(x-x_0)}} \quad \sigma(z)$$

$$L = 1, k = 1, x_0 = 0$$

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$



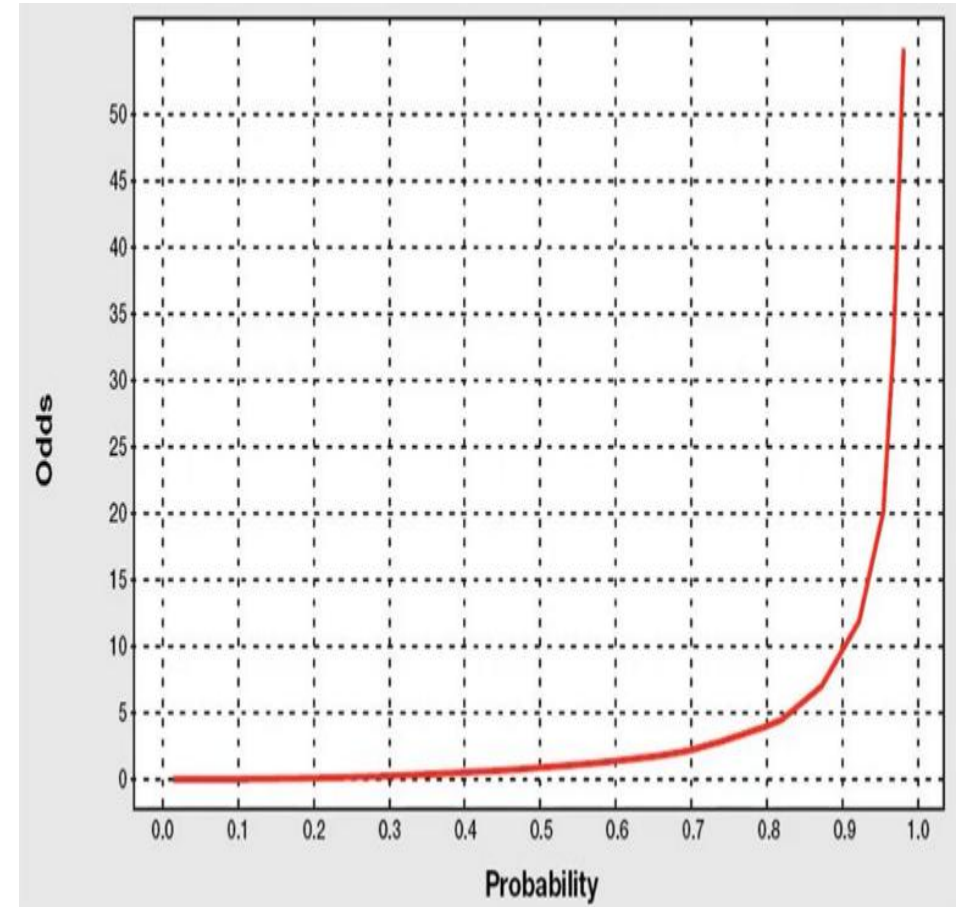
확률밀도함수(probability density function) 요건 충족

Odds

- 사건이 발생하지 않을 확률과 발생할 확률사이의 비

$$odds = \frac{P(A)}{P(A^c)} = \frac{P(A)}{1 - P(A)}$$

- A의 발생 확률이 커질수록 odds는 무한대 접근
- $0 < odds < \infty$



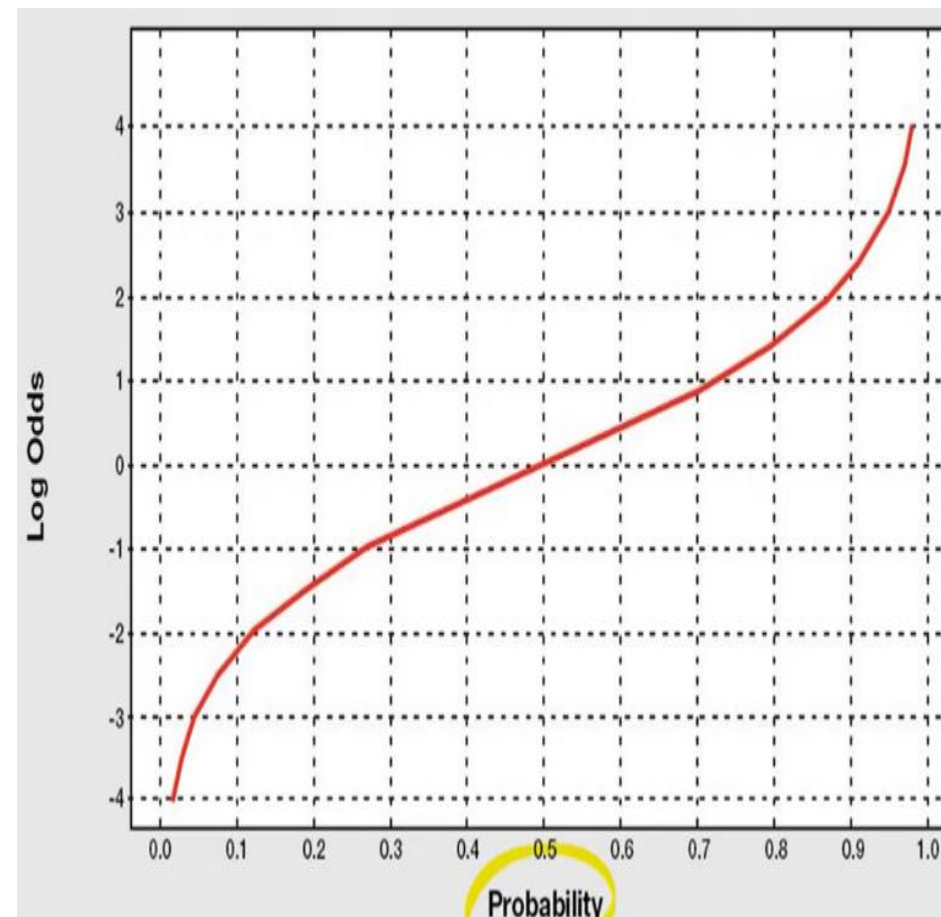
<https://ysyblog.tistory.com/177>

Logit

- Odds에 로그를 취한 값

$$\text{logit} = \ln(\text{odds}) = \ln\left(\frac{P(A)}{1 - P(A)}\right)$$

□ $-\infty < \text{logit} < \infty$



<https://ysyblog.tistory.com/177>

Logistic regression 유도

$$y = w_1x_1 + w_2x_2 + \cdots + w_nx_n + b = \vec{w}^T \vec{x}$$

좌변을 확률로 변경

$$P(Y = 1|X) = \vec{w}^T \vec{x}$$

좌변을 odds로 변경

$$\frac{P(Y = 1|X)}{1 - P(Y = 1|X)} = \vec{w}^T \vec{x}$$

좌변을 logit으로 변경

$$\ln \left(\frac{P(Y = 1|X)}{1 - P(Y = 1|X)} \right) = \vec{w}^T \vec{x} \longrightarrow P(Y = 1|X) = \frac{1}{1 + e^{-\vec{w}^T \vec{x}}}$$

양변에 지수를 취하고 정리

Logistic regression hypothesis

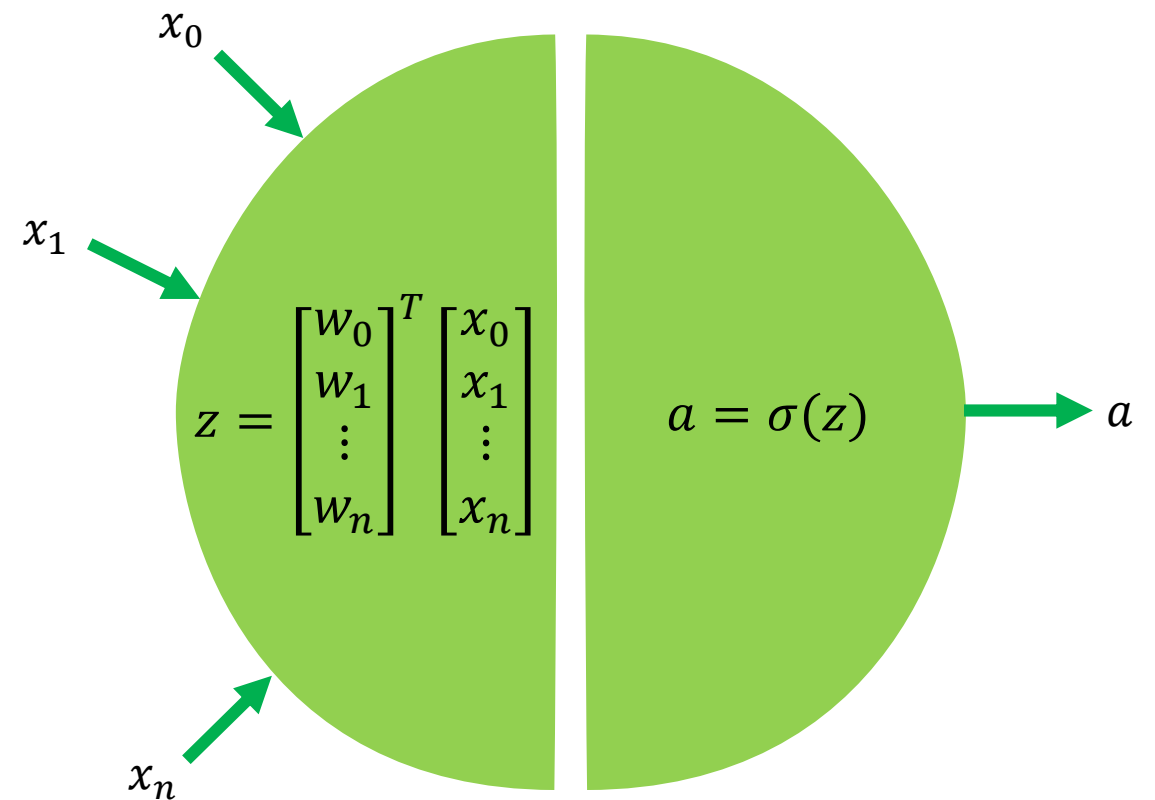
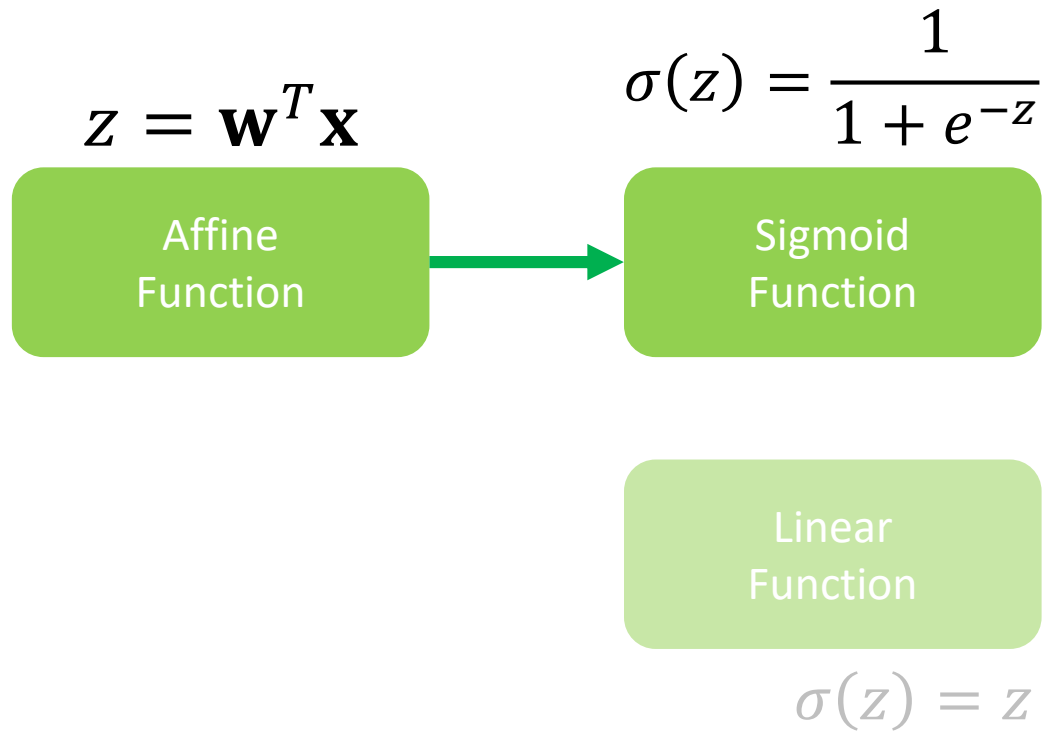
$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

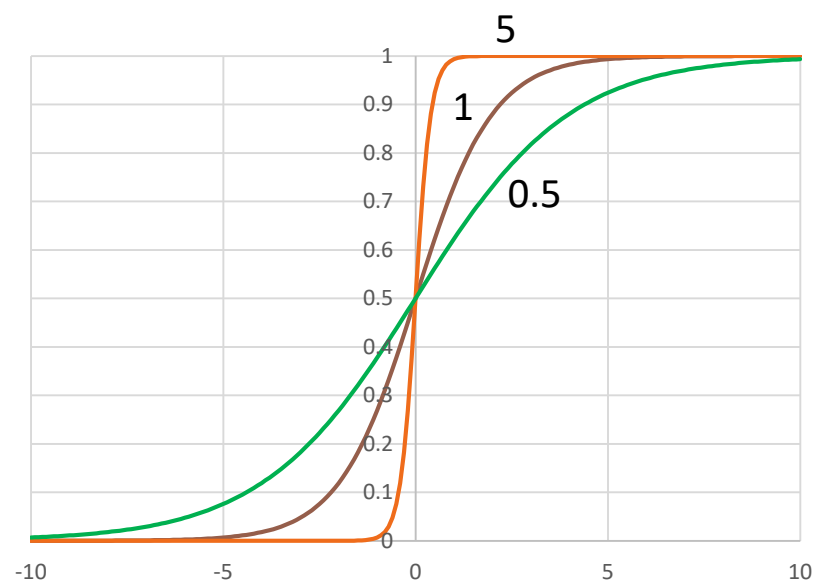
$$z = \mathbf{w}^T \mathbf{x}$$

$$H(\mathbf{x}) = \frac{1}{1 + e^{-\mathbf{w}^T \mathbf{x}}}$$

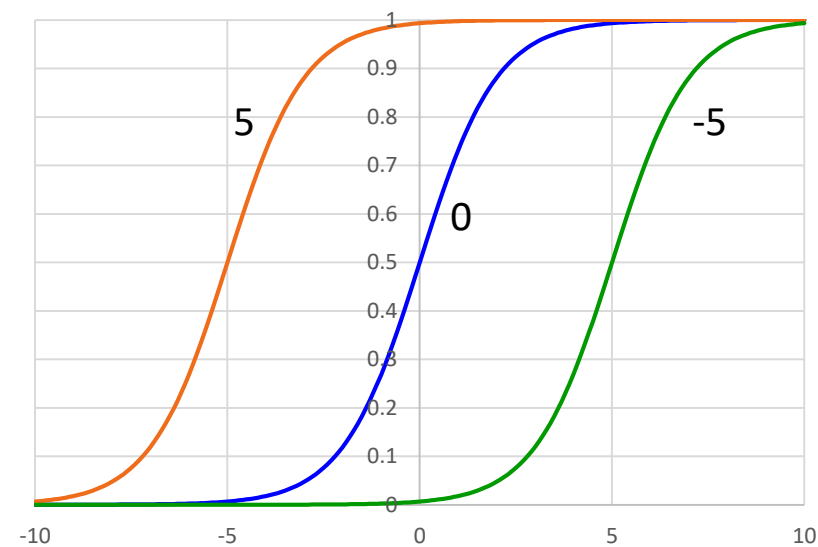
Logistic Regression: $0 < H(\mathbf{x}) < 1$

Affine and Sigmoid function





w 에 따른 그래프 ($b=0$)

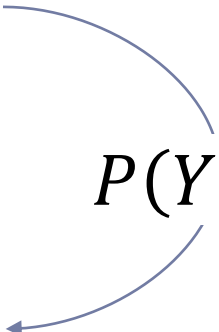


b 에 따른 그래프 ($w=1$)

결정경계 (decision boundary) (1)

$$P(Y = 1|X) = \frac{1}{1 + e^{-\vec{w}^T \vec{x}}}$$

$$P(Y = 0|X) = \frac{e^{-\vec{w}^T \vec{x}}}{1 + e^{-\vec{w}^T \vec{x}}}$$

$$P(Y = 1|X) + P(Y = 0|X) = 1$$


Positive 부류로 분류하기 위한 기준

$$P(Y = 1|X) > P(Y = 0|X)$$

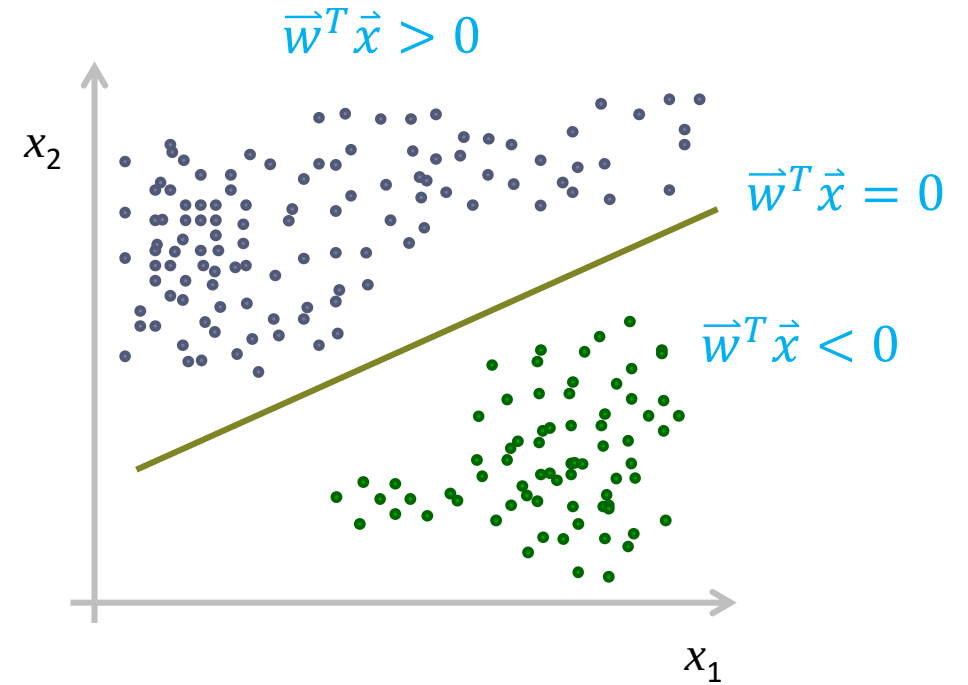
결정경계 (decision boundary) (2)

$$P(Y = 1|X) > P(Y = 0|X)$$

$$\frac{P(Y = 1|X)}{1 - P(Y = 1|X)} > 1$$

$$\ln \left(\frac{P(Y = 1|X)}{1 - P(Y = 1|X)} \right) > 0$$

$$\therefore \vec{w}^T \vec{x} > 0$$

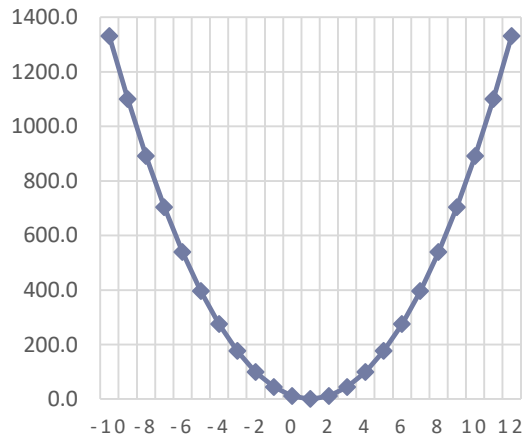


cost function

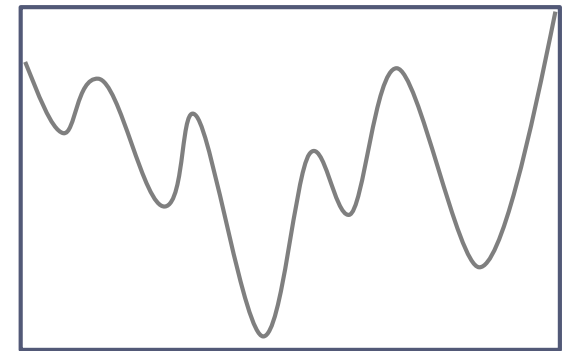
$$J(\mathbf{w}) = \frac{1}{n} \sum_{k=1}^n (H(x^{(k)}) - y^{(k)})^2$$

$$H(\mathbf{x}) = \mathbf{w}^T \mathbf{x}$$

$$H(\mathbf{x}) = \frac{1}{1 + e^{-\mathbf{w}^T \mathbf{x}}}$$

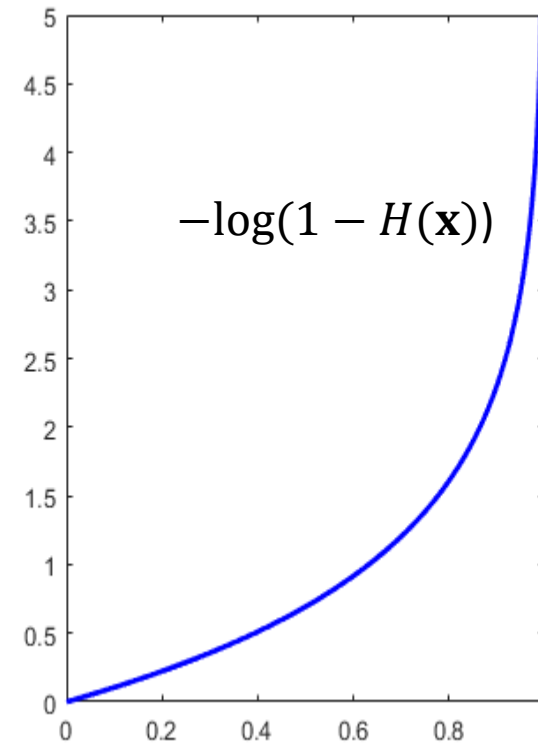
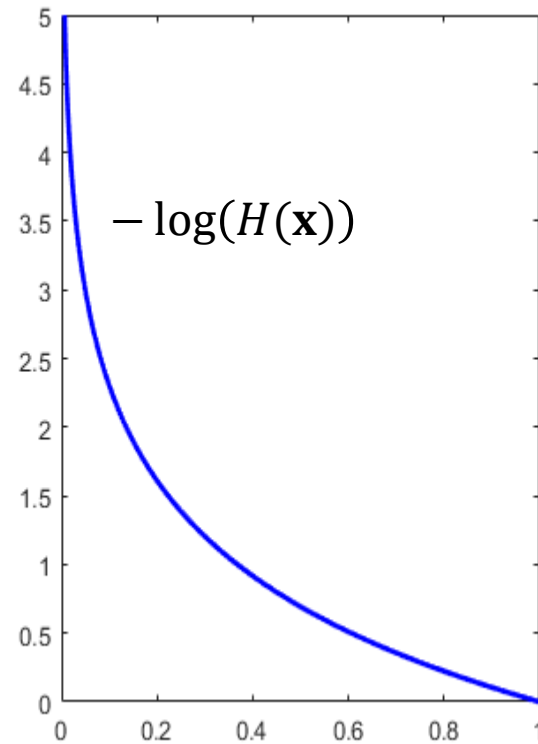


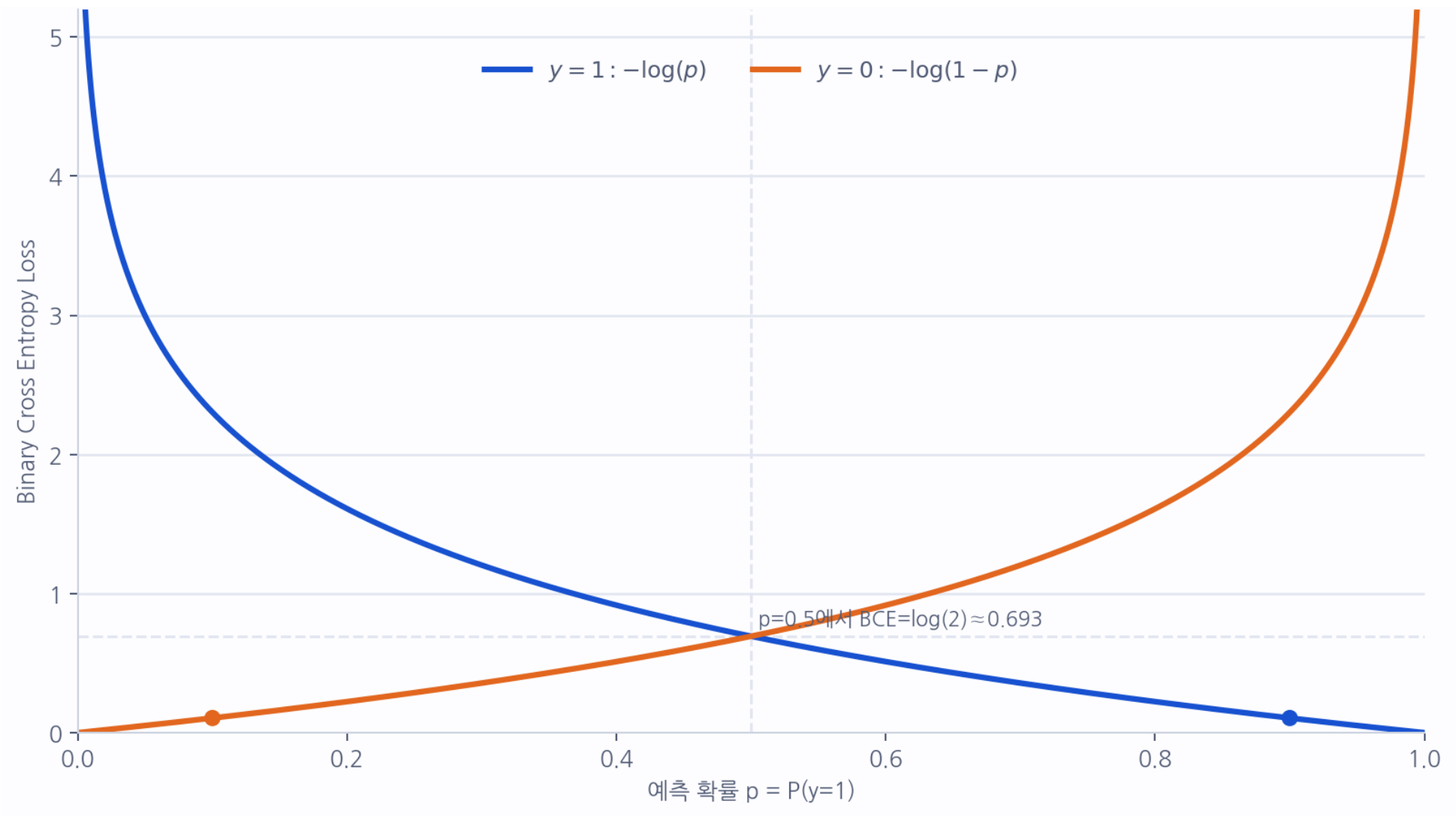
non-convex function



convex logistic regression cost function

$$\text{loss}(H(x)) = \begin{cases} -\log(H(\mathbf{x})), & \text{if } y = 1 \\ -\log(1 - H(\mathbf{x})), & \text{if } y = 0 \end{cases}$$





New cost function for logistic regression

$$\text{loss}(H(x)) = \begin{cases} -\log(H(\mathbf{x})), & \text{if } y = 1 \\ -\log(1 - H(\mathbf{x})), & \text{if } y = 0 \end{cases}$$

$$J(\mathbf{w}) = \frac{1}{n} \sum_{k=1}^n \left(-y^{(k)} \cdot \log(H(x^{(k)})) - (1 - y^{(k)}) \log(1 - H(x^{(k)})) \right)$$

Binary Cross Entropy (BCE)

Minimize cost

$$J(\mathbf{w}) = \frac{1}{n} \sum_{k=1}^n \left(-y^{(k)} \cdot \log(H(x^{(k)})) - (1 - y^{(k)}) \log(1 - H(x^{(k)})) \right)$$

Gradient descent:

$$\mathbf{w}^{(t+1)} := \mathbf{w}^{(t)} - \alpha \frac{\partial}{\partial w_i} J(\mathbf{w})$$

for $i = 0, 1, 2, \dots, m$

$$\frac{\partial}{\partial w_i} J(\mathbf{w}) = \frac{1}{n} \sum_{k=1}^n x^{(k)} (H(x^{(k)}) - y^{(k)})$$

Review

$$z = \mathbf{w}^T \mathbf{x}$$

Affine
Function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Logistic
Function

$$J(\mathbf{w}) = \frac{1}{n} \sum_{k=1}^n BCE^{(i)}$$

Cross Entropy
Loss

$$\mathbf{w} := \mathbf{w} - \alpha \frac{\partial}{\partial w_i} J(\mathbf{w})$$

Gradient Descent
Method

코딩으로 확인하는 로지스틱 회귀 (1)

- 015_01_Logistic_Regression.ipynb
- 015_02_Logistic_Regression_sklearn.ipynb

코드 비교

- 결정 경계가 $x = 7$ 근처에 있으면 모든 데이터를 정확하게 분류 가능

$$wx + b = 0 \Rightarrow x = -\frac{b}{w}$$

방법	w	b	결정 경계 $-b/w$
직접 구현 Batch GD	2.555	-17.732	약 6.94
직접 구현 SGD	3.778	-26.267	약 6.95
sklearn	0.961	-6.725	약 7.00

- 학습을 계속할수록 파라미터가 커지는 방향으로 진행
- `LogisticRegression( penalty='l2', C=1.0, solver='lbfgs' )`

참고: sigmoid 함수 미분

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\begin{aligned}\frac{d}{dz} \sigma(z) &= \frac{d}{dz} \left(\frac{1}{1 + e^{-z}} \right) \\&= \frac{d}{dz} (1 + e^{-z})^{-1} \\&= -1 \cdot (1 + e^{-z})^{-2} \cdot \frac{d}{dz} (1 + e^{-z}) \\&= -\frac{1}{(1 + e^{-z})^2} \cdot e^{-z}(-1) \\&= \frac{1 + e^{-z} - 1}{(1 + e^{-z})^2} \\&= \frac{1}{1 + e^{-z}} - \frac{1}{(1 + e^{-z})^2} \\&= \frac{1}{1 + e^{-z}} \left(1 - \frac{1}{1 + e^{-z}} \right) \\&= \sigma(z)(1 - \sigma(z))\end{aligned}$$

참고: 비용 함수에 대한 편미분 (w)

$$\frac{\partial J(w)}{\partial w} = \frac{\partial(-y \cdot \ln(\sigma(z)) - (1-y) \cdot \ln(1-\sigma(z)))}{\partial w}$$

$$= -y \cdot \frac{1}{\sigma(z)} \frac{\partial \sigma(z)}{\partial w} - (1-y) \cdot \frac{1}{1-\sigma(z)} \frac{\partial (1-\sigma(z))}{\partial w}$$

$$= -y \cdot \frac{\sigma(z)(1-\sigma(z))}{\sigma(z)} \cdot \frac{\partial z}{\partial w} - (1-y) \cdot \frac{-\sigma(z)(1-\sigma(z))}{1-\sigma(z)} \cdot \frac{\partial z}{\partial w}$$

$$= -y \cdot (1-\sigma(z)) \cdot x + (1-y) \cdot \sigma(z) \cdot x$$

$$= -xy + xy \cdot \sigma(z) + x \cdot \sigma(z) - xy \cdot \sigma(z)$$

$$= x(\sigma(z) - y)$$

$$\frac{\partial J(w)}{\partial b} = ?$$

$$\frac{\partial \sigma(z)}{\partial w} = \frac{\partial \sigma(z)}{\partial z} \cdot \frac{\partial z}{\partial w}$$

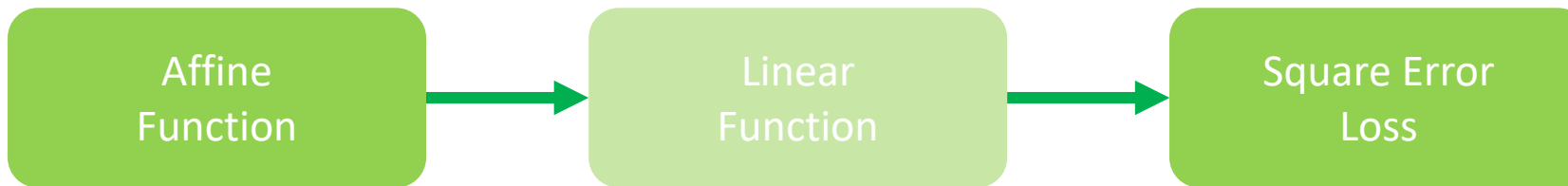
$$\frac{\partial (1-\sigma(z))}{\partial z} = -1 \cdot \frac{\partial \sigma(z)}{\partial z}$$

비용 함수를 b에 대해 편미분

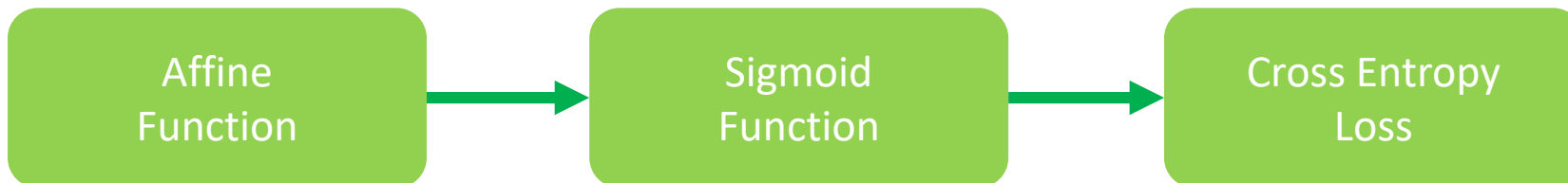
$$\begin{aligned}\frac{\partial J(w)}{\partial b} &= \frac{\partial(-y \cdot \ln(\sigma(z)) - (1 - y) \cdot \ln(1 - \sigma(z)))}{\partial b} \\&= -y \cdot \frac{1}{\sigma(z)} \frac{\partial \sigma(z)}{\partial b} - (1 - y) \cdot \frac{1}{1 - \sigma(z)} \frac{\partial (1 - \sigma(z))}{\partial b} \\&= -y \cdot \frac{\sigma(z)(1 - \sigma(z))}{\sigma(z)} \cdot \frac{\partial z}{\partial b} - (1 - y) \cdot \frac{-1 \cdot \sigma(z)(1 - \sigma(z))}{1 - \sigma(z)} \cdot \frac{\partial z}{\partial b} \\&= -y \cdot (1 - \sigma(z)) \cdot 1 + (1 - y) \cdot \sigma(z) \cdot 1 \\&= -y + y \cdot \sigma(z) + \sigma(z) - y \cdot \sigma(z) \\&= \sigma(z) - y\end{aligned}$$

Objects of Linear/Logistic Regression

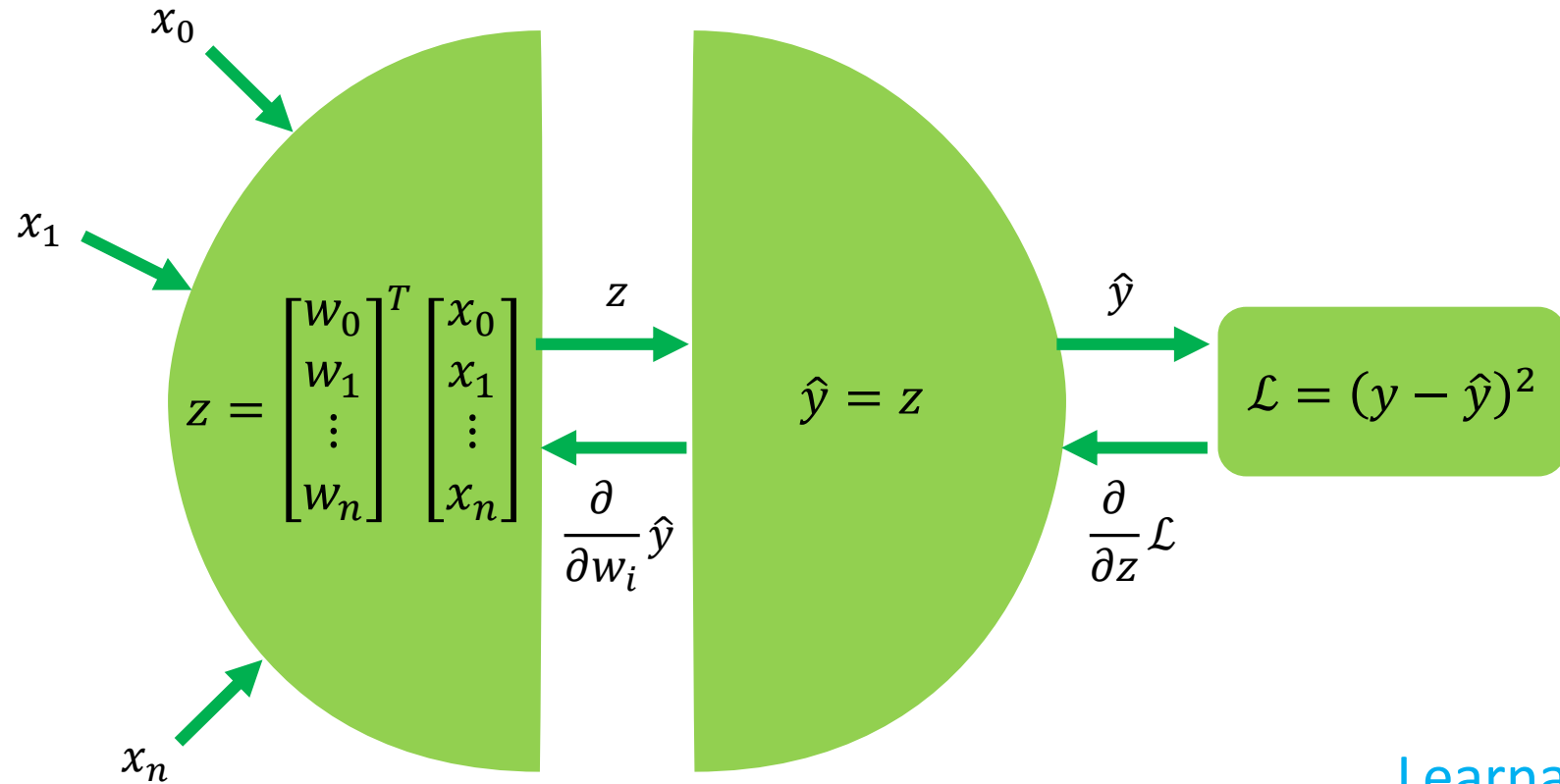
Linear Regression  **Regression**



Logistic Regression  **Binary Classification**

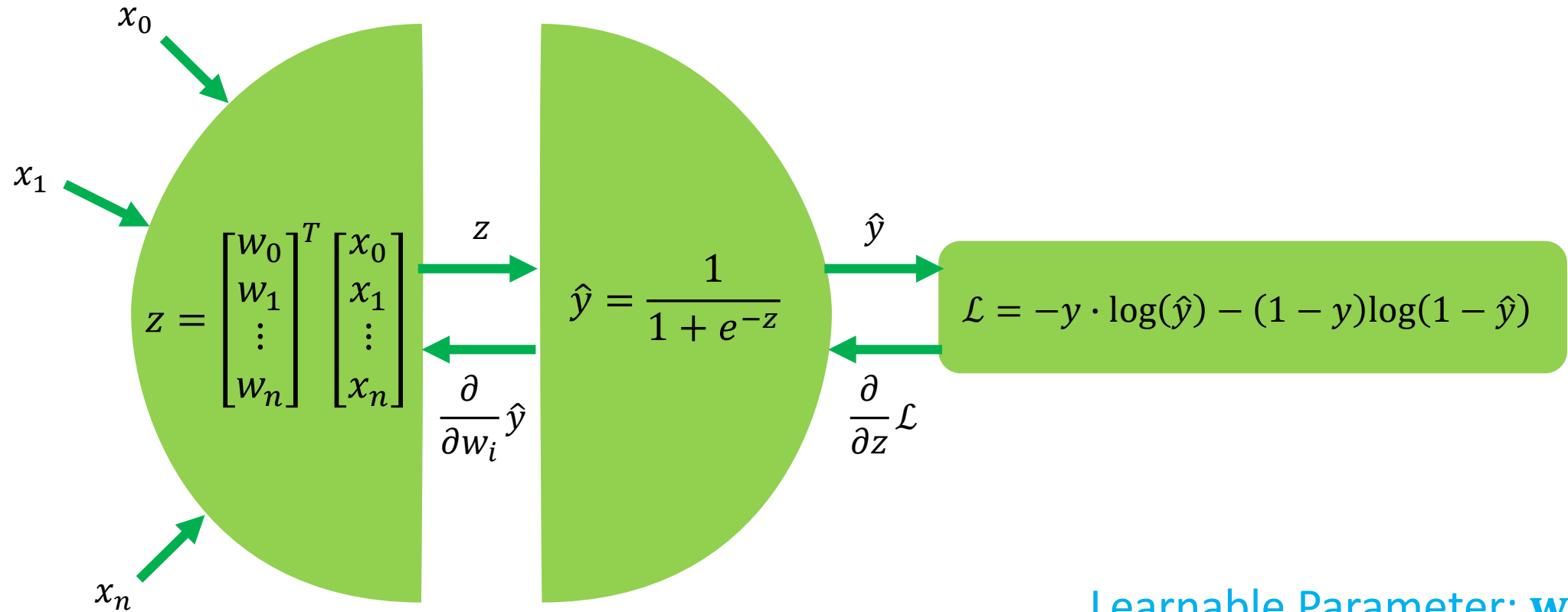


Learnable Parameters of Linear Regression



Learnable Parameter: $\mathbf{w}$

Learnable Parameters of Logistic Regression



코딩으로 확인하는 로지스틱 회귀 (2)

- 015_03_Linear_Regression_TF.ipynb
- 015_04_Logistic_Regression_TF.ipynb

참고자료

- 모두를 위한 머신러닝/딥러닝 (홍콩과기대 김성훈교수)

- <http://hunkim.github.io/ml/>

- Stanford Machine Learning

- <http://www.holehouse.org/mlclass/>

- Machine Learning (한성대 지준교수)

- <http://jun.hansung.ac.kr/ml>